ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PRACTICE EXAM 1

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 7 year: the design process, structures and forces, materials and processing, mechanical systems (levers, mechanical advantage and linkages) and electrical systems (circuits, conductors and electromagnets).

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings. Drawings must be neat and labelled.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 An egg shell is an example of a …

A frame structure. B shell structure. C solid structure. D mass structure.

1.2 A force that squashes or pushes a material together is …

A tension. B torsion. C compression. D shear.

1.3 Adding triangles (triangulation) to a structure makes it more …

A rigid. B heavy. C flexible. D colourful.

1.4 A wheelbarrow is an example of a …

A first-class lever. B second-class lever. C third-class lever. D linkage.

1.5 The point on which a lever turns is called the …

A load. B effort. C output. D fulcrum.

1.6 Which material is a good conductor of electricity?

A Copper B Rubber C Plastic D Wood

1.7 The component that opens and closes a circuit is a …

A cell. B bulb. C switch. D buzzer.

1.8 The first step of the design process is to …

A make. B evaluate. C investigate. D communicate.

1.9 A pair of tweezers is an example of a …

A third-class lever. B first-class lever. C second-class lever. D pulley.

1.10 A drawing that shows an object in 3D with its sides drawn at 30° to the horizontal is called …

A an orthographic drawing. B an oblique drawing.
C a floor plan. D an isometric drawing.

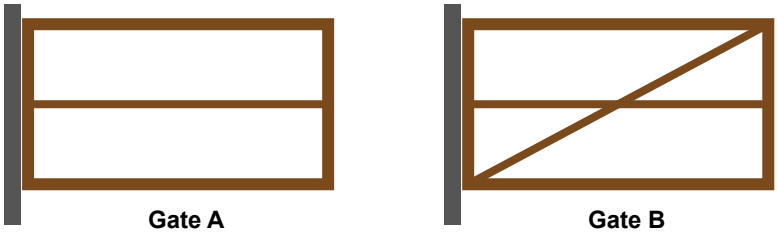
[10]

QUESTION 2: STRUCTURES

2.1 Classify each structure as a frame, shell or solid structure: (a) an electricity pylon; (b) a concrete dam wall; (c) a cool-drink can. (3)

2.2 Name the force that acts in each case: (a) the rope in a tug-of-war; (b) the legs of a chair when someone sits on it; (c) wringing out a wet cloth; (d) a diving board when a swimmer stands at the end. (4)

2.3 Study the two farm gates below.



Gate A **Gate B**

2.3.1 Which gate will stay in shape better over time? (1)

2.3.2 Explain your answer by referring to the shape formed by the extra bar. (2)

2.4 Name TWO other methods that can be used to make a structure stronger. (2)

2.5 Explain the difference between a static load and a dynamic load. Give ONE example of each on a bridge. (4)

2.6 Give TWO requirements of a well-designed structure. (2)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Give ONE property that makes each material suitable for its use: (a) glass for windows; (b) steel for bridges; (c) rubber for tyres; (d) cotton for T-shirts. (4)

3.2 (a) Give TWO functions of packaging. (b) Why should packaging be made from materials that can be recycled? (4)

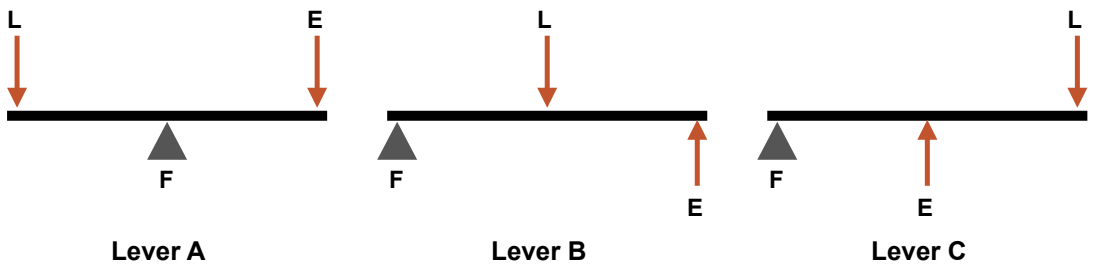
3.3 Name TWO methods of preserving food and explain how ONE of them stops food from going bad. (3)

3.4 Name a hand tool that is used to cut thin sheet metal. (1)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 Identify the class of each lever. E = effort, L = load, F = fulcrum.



Lever A **Lever B** **Lever C**

(3)

4.2 A lever lifts a load of 600 N with an effort of 200 N. Calculate the mechanical advantage. Use: MA = load ÷ effort. (2)

4.3 A crowbar has an effort arm of 1,2 m and a load arm of 0,3 m. (a) Calculate the mechanical advantage. Use: MA = effort arm ÷ load arm. (b) What effort is needed to lift a load of 800 N? (4)

4.4 Name the linkage that: (a) makes the output move in the opposite direction to the input; (b) changes the direction of motion through 90°. (2)

4.5 Give TWO examples of levers that are used in the kitchen. (2)

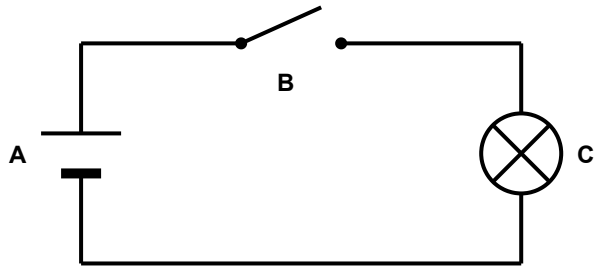
4.6 Draw a systems diagram (input → process → output) for a bicycle brake. (3)

4.7 Explain why it is easier to loosen a tight nut with a long spanner than with a short spanner. (2)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 Study the circuit diagram.



5.1.1 Name the components labelled A, B and C. (3)

5.1.2 Will component C work in this circuit? Explain your answer. (2)

5.2 Draw the circuit symbols for (a) a battery of two cells and (b) a buzzer. (2)

5.3 Classify each object as a conductor or an insulator: (a) a copper wire; (b) a plastic ruler; (c) a steel paper clip; (d) a rubber eraser. (4)

5.4 (a) Explain how you can make a simple electromagnet. (b) Give TWO ways to make an electromagnet stronger. (c) Give ONE use of an electromagnet. (5)

5.5 Give ONE safety rule when working with electricity at home. (1)

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TOTAL: 75

END OF QUESTION PAPER